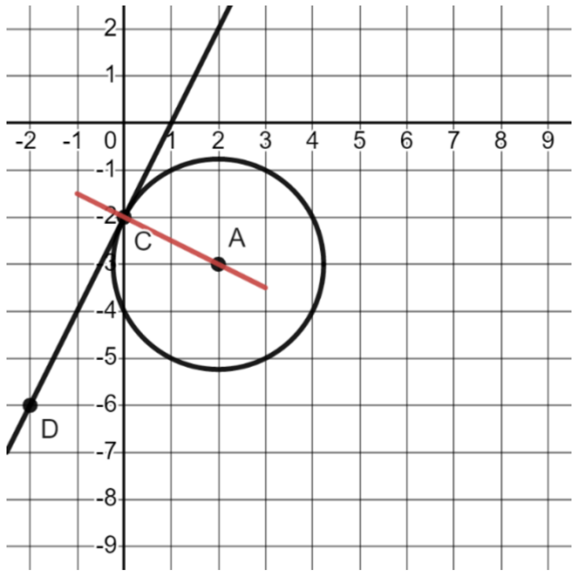
Geometry
Unit 13
Lesson 2 Practice

Name__Answer Key________
Date______________________
Period_____________________

Use the following information for questions 1-4. Circle A is shown, where point C is on the circle and line CD is tangent to circle A.



1. Use a straightedge to draw a line that goes through points A and C.

2. Write the equation of line CD.
Slope: 2
$y - y_1 = m(x - x_1)$
$y - (-6) = 2(x - (-2))$
$y + 6 = 2(x + 2)$
$y = 2x - 2$

3. Write the equation of line AC.
Slope: $-\frac{1}{2}$
$y - y_1 = m(x - x_1)$
$y - (-2) = -\frac{1}{2}(x - 0)$
$y + 2 = -\frac{1}{2}x$
$y = -\frac{1}{2}x - 2$

4. Line CD and line AC intersect at point $(0, -2)$. Use the equations of line CD and AC to confirm this point.
$y = 2x - 2$
$-2 = 2(0) - 2$
$-2 = -2$ ✓
$y = -\frac{1}{2}x - 2$
$-2 = -\frac{1}{2}(0) - 2$
$-2 = -2$ ✓

Use the following information for questions 5-8. Circle P has a center at $(4,-3)$ and a tangent line with the equation $y = 4x - 2$.

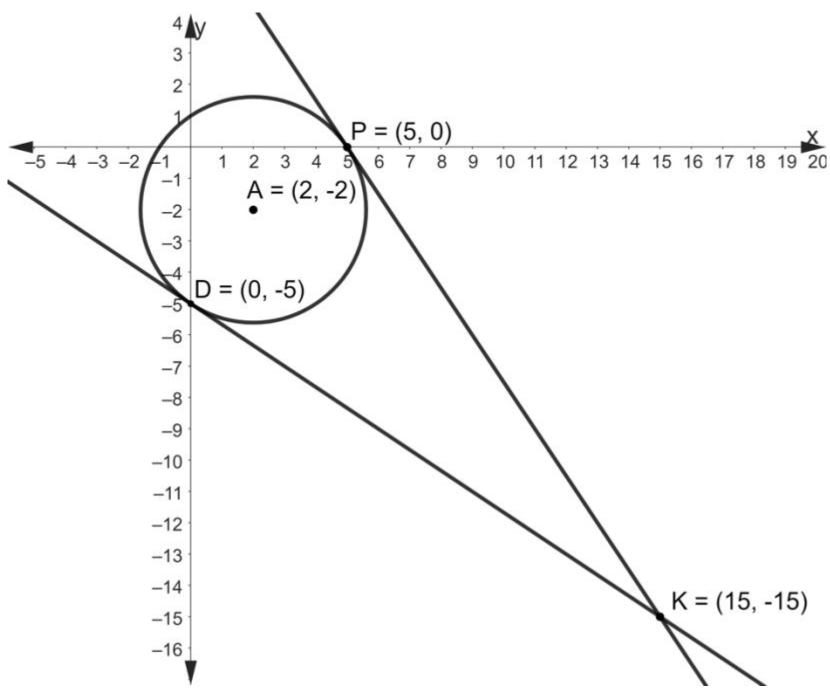
5. Write the equation of the line perpendicular to the tangent line.
Slope: $\frac{4}{1} \rightarrow \frac{1}{4} \rightarrow -\frac{1}{4}$
$y = -\frac{1}{4}x - 2$

6. Find the point where the tangent line touches circle P. Call the point H.
H is $(0,-2)$

7. What is the radius of circle P?
$r = \sqrt{17}$

8. Write the equation of circle P.
$(x-4)^2 + (y+3)^2 = 17$

Use the following information for questions 9-10. Circle A is shown with tangent lines PK and DK.



9. Complete the table.

Line Segment	Length
$\overline{PK}$	$\sqrt{325}$
$\overline{DK}$	$\sqrt{325}$
$\overline{AD}$	$\sqrt{13}$

$10^2 + 15^2 = (PK)^2$
$10^2 + 15^2 = (DK)^2$
$(-3)^2 + 2^2 = (AD)^2$

10. Write the equation of circle A.
$(x-2)^2 + (y+2)^2 = 13$